INF1002
Programming Fundamentals
Lecture 5: Recursion and Docstring

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Review

- File I/O
  - read, readline, readlines
  - write, writelines
  - with open() as file:
    for line in file
- Function
  - Default arguments
  - Positional and keyword arguments
  - Will a Variable's Value Change After a Function Call?
  - Variable scope
- Modules
- Higher-order functions
  - Function can be the value of a variable
  - Can also be passed and returned just like any other reference variables

Outline

- Packing and Unpacking
- Recursion
- Docstring

Tuple Packing & Unpacking

- In Python, a function can “return multiple values.”
- Actually, Python packs them into a tuple.
- You can then unpack the tuple into variables.
  - Unpacking assigns each element to separate variables.
  - Use _ for a single unused value.
  - Use *var to collect the “rest” of the values.
  - Use *_ to skip a block of values.

```python
def divide(a, b):
    return a // b, a % b, a / b

# Unpack into three variables
q, r, c = divide(10, 3)
print(q, r, c)        # 3 1 3.333...

# Use _ to ignore the third value
q, r, _ = divide(10, 3)
print(q, r)           # 3 1

# Use * to collect the remaining values
q, *y = divide(10, 3)
print(q, y)           # 3 [1, 3.333...]

# Use *_ to ignore all remaining values
q, *_ = divide(10, 3)
print(q)              # 3
```

```
3 1 3.3333333333333335
3 1
3 [1, 3.3333333333333335]
3
```

Multiple Assignment

- On the right
  - values are packed into a tuple.
- On the left
  - values are unpacked into variables.

```
first = 10
second = 20
print(first,second)
first,second = second,first
print(first,second)
```

```
10 20
20 10
```

List Comprehension*

- new_list = [expression for member in iterable]
- What is (expression for member in iterable)?

```
a = (x**2 for x in [1,2,3])
print(a)
for i in a:
  print(i)
```

```
<generator object <genexpr> at 0x7b52dd7ddd80>
1
4
9
```

List Comprehension*

- {expression for member in iterable}
  - set

- {member:exp(member) for member in iterable}
  - dictionary

Recursion

- In computer science, recursion is a method of solving a computational problem where the solution depends on solutions to smaller instances of the same problem.[1]
- Recursion occurs when the definition of a concept or process depends on a simpler or previous version of itself.[2]

1. https://en.wikipedia.org/wiki/Recursion_(computer_science)
2. https://en.wikipedia.org/wiki/Recursion
3. https://medium.com/analytics-vidhya/recursion-the-nesting-doll-of-programming-404ae61708a0

Factorial

- $n! = \begin{cases} 1, & n = 0 \\ n * (n-1)!, & n > 0 \end{cases}$
- n=3

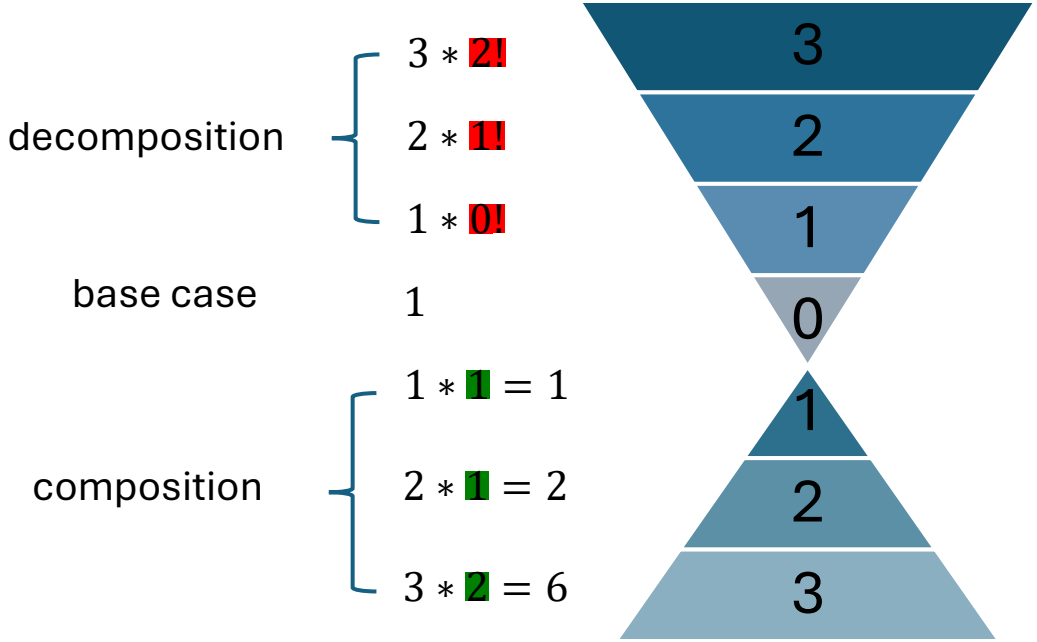
$$n! = 3 * (3-1)! = 3 * 2! = 3 * (2 * (2-1)!) = 3 * (2 * 1!) = 3 * (2 * (1 * (1-1)!)) = 3 * (2 * (1 * 0!)) = 6$$

How to we write a program to solve this problem?

Factorial

- decomposition
  - We turn this problem into a smaller problem of same kind.
- base case
  - Finally, we know the answer without calling the function itself
- composition
  - We use the result of smaller problems to find the result of a larger problem.

$$n! = \begin{cases} 1, & n = 0 \\ n * (n-1)!, & n > 0 \end{cases}$$



Factorial

- https://pythontutor.com/render.html#mode=display
- Further Reading:
  - Stack
    - Last in first out
    - Push and pop
  - Frame
    - To represent the execution state of a function or code block
      - Function Name
      - Local Variables
      - Global Variables
      - Instruction Pointer
      - Previous Frame

```
def factorial(n):
    # Base case: If n is 0, return 1
    if n == 0:
        return 1
    # Recursive case: n! = n * (n-1)!
    else:
        return n * factorial(n - 1)
# Example usage
number = int(input("Please enter an integer: "))
result = factorial(number)
print(f"The factorial of {number} is {result}")
```

Key points

- Figure out the formula about how a big problem can be solved by a **similar smaller problem**
  - Assume you know how to solve the problem for *n-1*. How to use this information to solve the problem for *n*?
  - Figure out what is the base case that the problem can get a solution.

Fibonacci numbers

- Leonardo Pisano Fibonacci (12thcentury) is credited for the sequence:
  - f(n): 0, 1, 1, 2, 3, 5, 8, 13, 21, ...
  - n: 0, 1, 2, 3, 4, 5, 6, 7, 8, ...
- Starting from the third number, each number in the sequence is the sum of the previous two.
- Calculate f(n) to calculate the number on the position n

Fibonacci numbers

- Figure out the formula about how a big problem can be solved by a similar smaller problem
  - Assume you know how to solve the problem for *n-1*. How to use this information to solve the problem for *n*? Derive the recursive formula.
  - Figure out what is the base case that the problem can get a solution.

$$F_0 = 0$$

- $$F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2}$$

Base case

Recursive formula

From analysis to program

```
def func(big_problem):
    if base_case:
        return value
    else:
        # recursive formula
        tmp = func(smaller_problem)
        return recursive_formula(tmp)
```

Practice

- Reverse a list
  - scores = [70,63,98,85,22]
  - Figure out the formula about how a big problem can be solved by a similar smaller problem
    - Assume you know how to solve the problem for n-1. How to use this information to solve the problem for n?
    - Figure out what is the base case that the problem can get a solution.

  - What is the base case?
  - What is the recursive formula?

https://leetcode.com/problems/reverse-linked-list/

Practice

```python
scores = [70,63,98,85,22]
def reverse(scores):
  if len(scores)==0:
    return []
  elif len(scores)==1:
    return scores
  else:
    return reverse(scores[1:])+[scores[0]]

print(reverse(scores))
```

```
[22, 85, 98, 63, 70]
```

Practice

- Implement Factorial function without recursion

$$n! = \begin{cases} 1, & n = 0 \\ n * (n-1)!, & n > 0 \end{cases}$$

- Recursion problems can be written in an iterative manner, and vice versa
- In actual programming, the choice between recursion and iteration often depends on the nature and context of the specific problem, as well as considerations of performance and readability.

Practice

```python
def factorial_loop(n):
  res = 1
  for i in range(1,n+1):
    res = res*i
  return res

print(factorial_loop(5))
```

Review

- Recursion
  - Figure out the formula about how a big problem can be solved by a similar smaller problem
  - Base case
- Further Reading
  - Dynamic Programming
  - Stack
  - Frame

Docstring

- A Docstring is a comment that appears as the first line in a new part of the code
- Used for help functions

```python
import math
help(math)
```

```
Help on built-in module math:

NAME
    math

DESCRIPTION
    This module provides access to the mathematical functions
    defined by the C standard.

FUNCTIONS
    acos(x, /)
        Return the arc cosine (measured in radians) of x.

        The result is between 0 and pi.

    acosh(x, /)
        Return the inverse hyperbolic cosine of x.

    asin(x, /)
        Return the arc sine (measured in radians) of x.

        The result is between -pi/2 and pi/2.

    asinh(x, /)
        Return the inverse hyperbolic sine of x.

    atan(x, /)
        Return the arc tangent (measured in radians) of x.

        The result is between -pi/2 and pi/2.

    atan2(y, x, /)
        Return the arc tangent (measured in radians) of y/x.

        Unlike atan(y/x), the signs of both x and y are considered.
```

Write your own module

- In the beginning of the module, write the module description surrounded by ''' or """
- In each function, write your comment in the first line with ''' or """

txt_processor.py

```python
'''
A module that processing text files
'''
def load_data(file_name):
    '''
    load txt lines from a file
    '''
    with open(file_name,'r') as file:
        data = file.readlines()
        return data

def process_data(data):
    '''
    data is a list of string
    for each string, there is a \\n at the end
    '''
    data = [i.strip()+': processed\n' for i in data]
    return data
```

Write your own module

test.py

```python
import txt_processor
help(txt_processor)
#from txt_processor import load_data, process_data
```

```
Help on module txt_processor:

NAME
    txt_processor - A module that processing text files

FUNCTIONS
    load_data(file_name)
        load txt lines from a file

    process_data(data)
        data is a list of string
        for each string, there is a \n at the end

FILE
```

Byte Data Type

- file.read(size)
- ASCII, Byte, bit, Unicode
- Beyond txt file: image, audio, video, npy

Byte Data Type

- The byte is a unit of digital information that most commonly consists of eight bits.
- A bit is 0 and 1.
- The bit is the most basic unit of information in computing and digital communication.
  - The bit represents a logical state with one of two possible values. These values are most commonly represented as either "1" or "0", but other representations such as true/false, yes/no, on/off, or +/− are also widely used.
- What is the value range of a byte?
  - $2^8 = 256$
  - 0 to 255 or -128 to 127

Memory usage of int and float in Python*

- Not fixed
- A surprise from Python
  - import sys
  - print(sys.getsizeof(1.0)) # 24
  - print(sys.getsizeof(1)) # 28
  - reference count, type information, extra padding or management overhead, etc.

Byte and ASCII code

- One byte represents 8 bits
- The original ASCII table is a 7-bit encoding scheme that defines 128 characters, numbered from 0 to 127.
- These characters include English uppercase and lowercase letters, digits, punctuation marks, and control characters.
- The extended ASCII table (characters numbered 128 to 255) varies across different platforms and systems, lacking a single standard version.

ASCII Table

- ASCII stands for American Standard Code for Information Interchange.

Dec	Hx	Oct	Char		Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL	(null)	32	20	040		Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH	(start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX	(start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX	(end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT	(end of transmission)	36	24	044	$	$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ	(enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK	(acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL	(bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS	(backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB	(horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF	(NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT	(vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF	(NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR	(carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO	(shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI	(shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE	(data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1	(device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2	(device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3	(device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4	(device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK	(negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN	(synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB	(end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN	(cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM	(end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB	(substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC	(escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS	(file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174			|
29	1D	035	GS	(group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS	(record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US	(unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177	&#127;	DEL

Source: www.LookupTables.com

ASCII Table

- Extended ASCII Codes
- Has not been widely or consistently adopted

128	Ç	144	É	160	á	176	░	192	└	208	╨	224	α	240	≡
129	ü	145	æ	161	í	177	▒	193	┴	209	╤	225	ß	241	±
130	é	146	Æ	162	ó	178	▓	194	┬	210	╥	226	Γ	242	≥
131	â	147	ô	163	ú	179	│	195	├	211	╙	227	π	243	≤
132	ä	148	ö	164	ñ	180	┤	196	─	212	╘	228	Σ	244	⌠
133	à	149	ò	165	Ñ	181	╡	197	┼	213	╒	229	σ	245	⌡
134	å	150	û	166	ª	182	╢	198	╞	214	╓	230	µ	246	÷
135	ç	151	ù	167	º	183	╖	199	╟	215	╫	231	τ	247	≈
136	ê	152	ÿ	168	¿	184	╕	200	╚	216	╪	232	Φ	248	°
137	ë	153	Ö	169	⌐	185	╣	201	╔	217	┘	233	Θ	249	∙
138	è	154	Ü	170	¬	186	║	202	╩	218	┌	234	Ω	250	·
139	ï	155	¢	171	½	187	╗	203	╦	219	█	235	δ	251	√
140	î	156	£	172	¼	188	╝	204	╠	220	▄	236	∞	252	ⁿ
141	ì	157	¥	173	¡	189	╜	205	═	221	▌	237	φ	253	²
142	Ä	158	₧	174	«	190	╛	206	╬	222	▐	238	ε	254	■
143	Å	159	ƒ	175	»	191	┐	207	╧	223	▀	239	∩	255	

Source: www.LookupTables.com

www.LookupTables.com

Unicode

- The original ASCII table is a 7-bit encoding scheme that defines 128 characters, numbered from 0 to 127.
- These characters include English uppercase and lowercase letters, digits, punctuation marks, and control characters.
- How to handle other languages? Chinese, Tamil, etc.
  - Too many characters
  - A-65
  - 龙-40857

Unicode Character “龙” (U+9F99)

龙

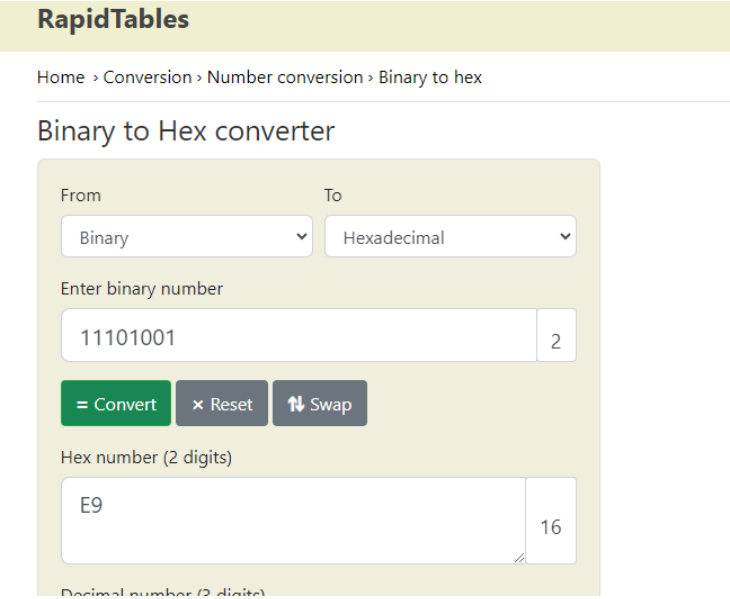
Name:	CJK Unified Ideograph-9F99 [1]
Unicode Version:	1.1 (June 1993) [2]
Block:	CJK Unified Ideographs, U+4E00 - U+9FFF [3]
Plane:	Basic Multilingual Plane, U+0000 - U+FFFF [3]
Script:	Han (Hanzi, Kanji, Hanja) (Hani) [4]
Category:	Other Letter (Lo) [1]
Bidirectional Class:	Left To Right (L) [1]
Combining Class:	Not Reordered (0) [1]
Character is Mirrored:	No [1]
HTML Entity:	龙 龙
UTF-8 Encoding:	0xE9 0xBE 0x99
UTF-16 Encoding:	0x9F99
UTF-32 Encoding:	0x00009F99

Unicode and UTF-8

- A – 65 – 1000001 – 0100 0001
- 龙 – 40857 – 1001 1111 1001 1001 – (1001 1111) (1001 1001)
- How can we know this is A龙? 0100 0001 1001 1111 1001 1001
  - (0000 0000 0100 0001) (1001 1111 1001 1001)
- UTF-8
  - Variable-Length Encoding
  - Uses 1 to 4 bytes to encode characters.
  - ASCII characters use 1 byte, while other characters use 2 to 4 bytes.
  - Full compatible with ASCII
    - Any valid ASCII text is also valid UTF-8 text.

Byte and UTF-8

- The character "龙" (U+9F99) is encoded in UTF-8 as E9 BE 99.
- 龙 – 40857 – 1001 1111 1001 1001 - 1001 111110 011001
- UTF-8 uses 3 bytes to represent "龙"
- Format: 1110xxxx 10xxxxxx 10xxxxxx
- First byte 1110xxxx
  - Fill in the first four bits: 1110 1001, which is E9.
- Second byte 10xxxxxx
  - Fill in the next six bits: 1011 1110, which is BE.
- Third byte 10xxxxxx:
  - Fill in the remaining six bits: 1001 1001, which is 99.



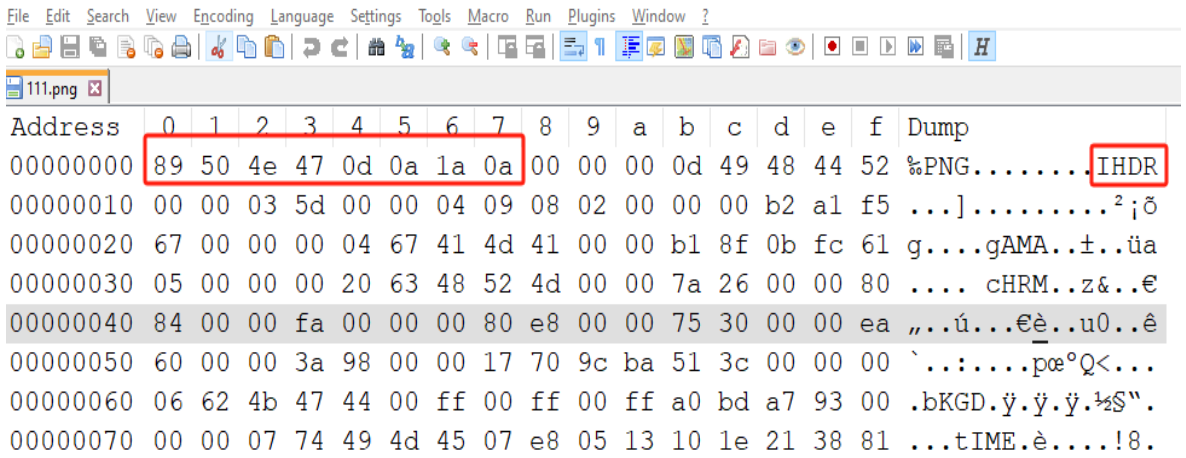
https://www.rapidtables.com/convert/number/binary-to-decimal.html

UTF-8

Type	Description	Unicode range	Format
1 Byte	Encode 7-bit ASCII characters	U+0000 to U+007F	0xxxxxxx
2 Bytes	Encode 8 to 11-bit characters	U+0080 to U+07FF	110xxxxx 10xxxxxx
3 Bytes	Encode 12- to 16-bit characters	U+0800 to U+FFFF	1110xxxx 10xxxxxx 10xxxxxx
4 Bytes	Encode 17- to 21-bit characters	U+10000 to U+10FFFF	11110xxx 10xxxxxx 10xxxxxx 10xxxxxx

Understanding binary files

- Structure of a PNG File
  - Signature
    - First 8 bytes of the file
    - Specific byte values (hexadecimal)
      - 89 50 4E 47 0D 0A 1A 0A
    - Indicates that the file is a PNG
  - Chunks
    - Critical chunks (must appear in the file)
      - IHDR: Image header
        - Width,Height, Bit depth, Color type, ...
      - IDAT: Image data
      - IEND: Image end
    - Ancillary chunks (optional metadata)



Notepad++
Install HexViewer plugin

Bits, Bytes, and Character Encoding

- **Bit**:
  - **1 bit** represents the most basic unit of data in computing, either **0** or **1**.
- **Byte**:
  - A **byte** is the **standard unit** for storing data in computers, comprising **8 bits**.
  - A byte can represent 256 possible values (from 0 to 255 in decimal).
- **ASCII** (American Standard Code for Information Interchange):
  - ASCII uses **7 bits** to encode printable characters, including English letters, digits, and control characters.
  - Typically, ASCII is represented using **8 bits** (1 byte), where the highest bit is often set to 0.
- **UTF-8** (Unicode Transformation Format - 8 bits):
  - UTF-8 is a **variable-length** encoding scheme that uses **1 to 4 bytes** to represent characters from all languages.
  - It is backward compatible with ASCII: ASCII characters use 1 byte, while more complex characters (e.g., from other languages) can use up to 4 bytes.
- **Binary Files**:
  - A **binary file** is any file that contains a sequence of bytes.
  - Not all binary files are **human-readable**. Text files, such as those using **ASCII** or **UTF-8** encoding, can be read by humans. However, binary files like images, audio, or executables often need to be **decoded** by specific programs to be interpreted in a human-readable form.

UTF-8 won't be on the exam.